SMART-STUA: A MOBILE APPLICATION INTERNET OF THINGS SYSTEM FOR AFLATOXIN CONTAMINATION MONITORING AND PREVENTION IN GRAIN STORAGE

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A PROPOSAL SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE FOR THE STUDY LEADING TO A PROJECT REPORT IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE OF GULU UNIVERSITY.

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Approval Statement

By signing below, the undersigned acknowledges reviewing the attached project proposal and approves the execution of the project as described. This approval authorizes the allocation of resources and funds necessary to proceed to the initiation phase.

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1. INTRODUCTION

1.1. Background

Aflatoxins are potent, carcinogenic toxins produced by molds such as Aspergillus flavus, which thrive in the high-temperature and high-humidity conditions common in tropical regions. Classified as Group 1 human carcinogens, these toxins contaminate staple grains, leading to severe public health crises, including liver cancer and stunted growth in children, as well as massive economic losses from rejected agricultural exports. Aflatoxin contamination represents a critical global food safety and security challenge. Worldwide, it is estimated that over 25% of the global food crop is affected by mycotoxins annually, with aflatoxins being among the most toxic (Eskola et al., 2020). This contamination causes staggering economic losses, estimated in the billions of US dollars each year due to crop loss, impacted livestock health, and costs associated with regulatory monitoring and disease treatment.

The problem is disproportionately severe in Africa, where climatic conditions and post-harvest handling practices create a high-risk environment. The African Union estimates that Africa loses over US$670 million in annual trade due to aflatoxin contamination, severely limiting export potential. In terms of health, a meta-analysis published in the World Mycotoxin Journal indicated that 40% of grain samples in Africa exceed the stringent European Union regulatory limits for aflatoxins (Ndwata, Rashid, & Chaula, 2022). This widespread contamination contributes to high rates of liver cancer in parts of West Africa and is a chronic public health threat across the continent.

The situation in Uganda is particularly acute. Studies have shown that aflatoxin contamination is pervasive in staple foods like maize and groundnuts. A national survey conducted by the Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) found that up to 60% of maize samples in some Ugandan markets contained aflatoxin levels above the permissible limits (Akullo et al., 2023). This high prevalence directly impacts both health and the economy. For a population where maize is a staple food, this translates to continuous exposure. Economically, it undermines market access; for instance, Uganda faces recurring rejections of its grain exports by neighboring countries due to high aflatoxin levels, crippling farmer incomes and national revenue. This persistent problem underscores the urgent need for innovative, accessible, and proactive solutions to break the cycle of contamination at the storage level.

Current interventions, such as traditional sun-drying and manual inspection, are fundamentally reactive and inadequate. These methods cannot provide real-time data on storage conditions, leaving grain vulnerable to invisible mold growth until it is too late for effective intervention (Wanyonyi, 2025). Manual checks are infrequent, labor-intensive, and prone to human error, while sun-drying is inconsistent and weather dependent. Chemical treatments, though sometimes used, introduce risks regarding food safety and chemical residues. This technological gap perpetuates a cycle of crop loss, health endangerment, and economic damage for farmers and communities.

Recent research highlights the potential of Internet of Things (IoT) technologies to address this gap. Studies have demonstrated the effectiveness of IoT-based systems for real-time monitoring of grain storage parameters, leading to improved efficiency and reduced spoilage (Hema, Velmurugan, Sunil, Aziz, & Thirunavkarasu, 2020; Basciftci, Unlu, & Dasdemir, 2021). However, despite these advancements, the adoption of such technology in Uganda and similar developing regions remains negligible. The primary barriers include the high initial cost of existing systems, the technical expertise required for their operation, and a lack of integration with automated preventive mechanisms. Most existing systems focus solely on monitoring and alerting, still requiring a manual response to mitigate the detected risk.

To bridge this critical gap, this project proposes Smart-Stua, an integrated mobile app Internet of Things (IoT) system designed specifically for the context of small to medium scale grain storage in Uganda. Smart-Stua will use in-store sensors to continuously track temperature and humidity, transmit data via a long-range, low-power network (LoRaWAN), and utilize a cloud-based algorithm to calculate an Aflatoxin Risk Index (ARI) . By providing a user-friendly web dashboard with real-time alerts and, uniquely, by integrating the capacity to automatically trigger corrective mechanisms (such as a grain-drying system), Smart-Stua will transform grain storage management from a reactive process to a truly preventive one, ensuring both food safety and economic stability.

1.2. Problem Statement

In Uganda and similar tropical regions, grain storage is highly vulnerable to contamination by aflatoxins, potent carcinogenic toxins produced by molds. Current mitigation strategies, such as manual inspection, sun-drying, and chemical treatments, are fundamentally inadequate. Manual checks are infrequent and fail to capture real-time data, sun-drying is often inconsistent and weather-dependent (Wanyonyi, 2025), while chemical use raises safety concerns. These reactive methods are deployed after the problem has already occurred, leading to widespread spoilage. The result has severe consequences, including substantial economic losses for farmers from rejected produce, and grave public health risks for consumers, such as liver cancer and impaired child development.

The core of the problem is the absence of a proactive, real-time monitoring and preventive mechanism within storage facilities. Existing systems lack the integration of IoT sensors, data analytics, and responses necessary for prevention (Basciftci et al., 2021; Doshi, Patel, & Bharti, 2019). Without continuous, accurate data on critical environmental conditions like temperature and humidity, storage managers are left without the actionable insights needed to intervene before contamination starts. This critical technological gap perpetuates cycles of food insecurity, economic damage, and preventable health crises, underscoring the urgent need for an affordable, smart IoT-based solution to transform grain storage management from reactive to preventive. The study proposes a mobile app Internet of Things system for aflatoxin contamination monitoring and prevention in grain storage.

1.3. Objectives

1.3.1. Main Objective

To develop a mobile app Internet of Things system for aflatoxin contamination monitoring and prevention in grain storage.

1.3.2. Specific Objectives

i. To determine the requirements needed for designing the mobile app Internet of Things system for aflatoxin contamination monitoring and prevention in grain storage.

ii. To design a mobile app Internet of Things system for aflatoxin contamination monitoring and prevention in grain storage.

iii. To implement the mobile app Internet of Things system for aflatoxin contamination monitoring and prevention in grain storage.

iv. To test and validate the performance and efficiency of the system.

1.4. Scope

The study will focus on the design and development of a mobile app IoT system for monitoring temperature and humidity levels in grain storage facilities within Gulu District. The system will incorporate sensor nodes, cloud-based data processing, an Aflatoxin Risk

Index algorithm, and a mobile app dashboard for visualization and alerts. The project will not involve chemical analysis of aflatoxin levels but will rely on predictive environmental indicators.

1.5. Significance of the study

The proposed Smart-Stua system is significant in several ways:

i. Academically: It contributes to the body of knowledge in IoT-based agricultural monitoring systems and predictive risk modeling, specifically within the Ugandan context.

ii. Practically: It offers a technological solution to aflatoxin contamination, improving food safety and reducing post-harvest losses for farmers and storage cooperatives.

iii. Socio-economically: The system can enhance farmer incomes by reducing grain spoilage and preventing export rejection, while simultaneously protecting public health through safer food storage practices.

2. LITERATURE REVIEW

2.1. Overview of aflatoxin contamination

Aflatoxins are among the most dangerous mycotoxins affecting food crops worldwide. Studies indicate that climatic conditions and poor post-harvest handling significantly contribute to contamination in developing countries (Eskola et al., 2020). The problem is disproportionately severe in Africa with the situation being particularly acute in Uganda where studies have shown that alfatoxin contamination is pervasive in staple foods like maize and ground nuts (Ndwata et al., 2022). This translates into continuous exposure.

2.2. State-of-Art

Technological advancement and research in grain storage and mycotoxin prevention has aided in transitioning from traditional, reactive storage monitoring toward proactive, data-driven systems. These advancements focus on early detection of biological activity, allowing for interventions long before physical spoilage or aflatoxin accumulation becomes visible. These related systems include:

2.2.1. Integrated CO_2 monitoring system

The system employs an integrated approach to manage stored grain quality using carbon dioxide (CO_2) monitoring. It establishes that traditional methods relying on temperature and moisture content are insufficient for early spoilage detection, as they are often reactive. The core premise is that CO_2 generation from the respiration of grain, molds, and insects serves as a direct and early indicator of biological activity and spoilage. The paper reviews empirical models linking CO_2 production to storage conditions (temperature, moisture, time) for various grains. Critically, it discusses the physics of CO_2 diffusion through bulk grain, noting that its movement with convective air currents allows a single sensor in the headspace to detect spoilage hotspots located deep within the bin, a significant advantage over localized temperature cables. The review also surveys commercial CO_2 sensor technologies, predominantly non-dispersive infrared (NDIR) sensors, and advocates for a holistic monitoring system that integrates CO_2 data with temperature and relative humidity readings for proactive grain management.(Ramachandran & Singh, 2022).

2.2.2. Digital Twin (DT) framework

This introduces the concept of "sensorless monitoring," aiming to create a high-fidelity virtual model of a grain bin using minimal, farmer-accessible input data (e.g., mass added, moisture content, bin dimensions) to enable predictive, rather than reactive, management. The research is comprehensive, beginning with theoretical foundations in cybernetics and a review of pressure models like Janssen's equation. A key contribution is the experimental validation of a novel mathematical model that describes radial stress variation within grain. This was achieved using a custom-designed and calibrated tri-axial in-situ pressure sensor, which revealed that the lateral-to-vertical pressure ratio (k) varies significantly across the bin's radius. This pressure model was then integrated into a functional DT web application. The DT features layer-by-layer inventory tracking, visualizations of moisture and pressure distribution, and a prototype drying simulation. A significant finding from this simulation is that accounting for grain compaction, derived from the pressure model, substantially alters drying predictions, demonstrating the value of integrating these complex physical models into a practical management tool. The thesis concludes by discussing the platform's limitations and its broader implications for food traceability and the future of digital agriculture (Dyck, 2025).

2.3. State-of-Practice

Globally, grain preservation has shifted from manual monitoring toward integrated, technology-driven solutions. Modern systems increasingly utilize Internet of Things (IoT) frameworks to monitor environmental variables like temperature, relative humidity, and carbon dioxide (CO_2) levels. Unlike traditional methods that are often reactive, global practices now emphasize early detection of biological activity such as mold respiration before physical spoilage becomes visible. Advanced commercial and research models, such as those utilizing Digital Twin (DT) technology, provide real-time data visualization and predictive modeling to manage grain health throughout the storage lifecycle.

In Uganda, grain storage remains largely dominated by traditional post-harvest handling techniques, which contribute to high contamination rates. Most farmers rely on sun-drying on the ground and manual inspections (such as biting or touching) to assess dryness, methods that are inadequate for detecting invisible aflatoxin-producing molds. While there is a growing push by organizations like Uganda National Bureau of Standards (UNBS) and the Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) for improved practices (e.g., using tarpaulins and moisture meters), the adoption of real-time monitoring remains low. Current local efforts are primarily focused on awareness campaigns and the introduction of improved physical storage structures, such as hermetic bags and metal stores, rather than integrated digital monitoring systems.

Several systems exist that share functional similarities with the proposed Smart-Stua. Globally, Integrated CO_2 Monitoring Systems utilize gas sensors to detect the earliest signs of grain spoilage, offering a more sensitive alternative to simple temperature probes (Ramachandran & Singh, 2022). Additionally, Digital Twin Frameworks have been developed as proof-of-concepts to track grain properties layer-by-layer and simulate drying processes (Dyck, 2025). Locally and regionally, research prototypes of IoT-based real-time control systems are emerging; these use sensors and cloud connectivity to provide food grain procurement and storage data (Wanyonyi, 2025). However, many of these existing systems face challenges such as high computational requirements, complex user interfaces that are not farmer-friendly, and a lack of automated physical interventions, creating a gap that the Smart-Stua aims to fill.

2.3.1. Strengths and Weaknesses of Current Systems

System	Theory Used	Strength	Weakness	Gap Solved by the Proposed System
Integrated CO_2 Monitoring System	Resipration of grain, molds, and insects produces CO_2, making it an early indicator of biological activity and spoilage.	- Early spoilage detection (before temperature changes). - Gas mobility allows a single sensor to detect hotspots deep within the grain bulk. - Strong correlation with mycotoxin risk. - Based on established empirical models.	- Reactive (detects spoilage as it happens, not predictive). - Not toxin-specific (cannot distinguish toxigenic fungi or measure aflatoxins directly). - Higher cost and complexity than basic sensors.	The proposed system will bridge the gap between reactive detection and proactive management by using the sensor data not just for alerts, but as an input for a predictive model that helps make educated decissions on aflatoxin risk.
Digital Twin (DT) Framework	Physics based models (e.g., Janssen's equation for pressure, heat and mass transfer) to create a high-fidelity virtual replica of the grain bin's physical state.	- Predictive (simulates future states and risk). - High-fidelity modeling of physical processes (pressure, compaction, drying). - Sensorless monitoring philosophy minimizes sensor reliance. - Layer-by-layer tracking for precise management. - High integration potential with other farm systems.	- High complexity and not yet user-friendly for farmers. - A research prototype, not validated in diverse real-world conditions. - Limited biological modeling (does not model fungal growth or mycotoxin production). - Significant computational requirements.	The proposed system will address the DT's lack of real-time insight by integrating direct DHT sensing. This will ground the predictive physical model with actual data on conditions, creating a more holistic system that monitors both the physical state and the real-time biological risks like aflatoxin contamination.

2.3.2. The proposed system

Smart-Stua is designed as an integrated, end-to-end system that transforms traditional grain storage from a reactive process into a preventive one. Its architecture is built on four core layers that work in harmony: Physical Sensing, Connectivity, Cloud Intelligence, and User Interaction.

2.3.2.1. System Architecture

i. The Sensing Layer (In-store Hardware): At the heart of the system are distributed sensor nodes placed inside grain stores. Each node, built around a microcontroller (like an ESP32), is equipped with temperature and humidity sensors (such as the DHT22). These nodes are responsible for continuously capturing the environmental conditions of the stored grain.

ii. The Connectivity Layer (Data Transmission): The sensor nodes form a network and transmit data using a low-power, long-range wireless protocol like LoRaWAN. This data is received by a central LoRaWAN gateway, which acts as a bridge, aggregating information from all sensors in the storage facility and forwarding it to the cloud.

iii. The Cloud & Processing Layer (The Brain): This is the system's central intelligence unit. Built using a framework like Django, the cloud backend performs several critical functions:

    - It securely stores all incoming sensor data in a database.
    - It hosts the core Aflatoxin Risk Index (ARI) algorithm. This Python-based program analyzes the incoming temperature and humidity data against pre-established scientific models to calculate a real-time contamination risk level.
    - It manages the logic for triggering alerts and automated responses.

iv. The Application Layer (User Interface): The end-user interacts with the system through a mobile app dashboard. Designed for clarity and ease of use, the dashboard provides:

    - Real-time visualizations of temperature, humidity, and the calculated ARI.
    - Historical data trends and reports.
    - An interface for managing and receiving automated alerts (via SMS/Email).

2.3.2.2. How It Will Work: A Step-by-Step Operational Flow

The system's operation is a continuous, automated cycle of monitoring and analysis:

i. Continuous Monitoring: The in-store sensor nodes wake up at regular intervals to measure the ambient temperature and humidity of the grain mass. These readings are then transmitted wirelessly via the LoRaWAN gateway to the cloud platform.

ii. Risk Calculation & Analysis: The cloud platform receives the data and immediately feeds it into the Aflatoxin Risk Index (ARI) algorithm. This algorithm analyzes the current conditions and their duration to determine if they are conducive to the growth of aflatoxin-producing molds (e.g., Aspergillus flavus).

iii. Preventive Alerting:

- If the ARI exceeds a pre-defined safety threshold, the system identifies a developing risk.
- It instantly sends an automated alert (SMS/Email) to the store manager, notifying them of the danger.
- Crucially, the system is designed to go beyond mere alerts. As a preventive measure, it can automatically trigger connected correction mechanisms, such as an automated grain drying system, to reduce moisture and temperature without requiring any manual intervention.

iv. Visualization & Oversight: Throughout this process, the store manager can log into their mobile app dashboard at any time. They can monitor the live conditions, view the ARI trend over time.

3. Methodology

3.1. Overview of methodology to be used in this study

The methodology adopts an Agile approach within the System Development Life Cycle (SDLC) to ensure continuous improvement and adaptability through iterative development. The process is structured into four key phases: first, requirements gathering through direct stakeholder engagement in Gulu district to define functional and non-functional needs; second, system design involving architectural blueprints, database schemas, and user interface wireframes; third, development and implementation executed through focused sprints covering sensor node assembly, communication layer setup, cloud backend development with the Aflatoxin Risk Index algorithm, and web dashboard integration; and finally, testing and validation encompassing unit, integration, functional, and environmental tests, culminating in a pilot deployment to assess real-world performance and gather user feedback for refinement.

3.2. Adopted Agile methodology

Although there are various methodologies such as Waterfall, Dev Ops and Spiral, the project will follow the Agile methodology within the System Development Life Cycle (SDLC) framework, which will ensure continuous quality improvement through iterative development and frequent review as well as greater adaptability to changing requirements (Pargaonkar, 2023). Agile enables ongoing adjustment and improvement of the product as new user requirements emerge during data collection (Sinha & Das, 2021). Agile methodology will be applied in this study as follows;

3.2.1. Phase one: Requirements Gathering

Requirements will be collected from within Gulu district by directly engaging with grain storage operators, cooperative managers, and farmers through interviews, focus group discussions, questionnaires, and observations of current storage practices. The information gathered will help us derive functional and non-functional requirements for developing the system. Functional requirements will include real-time monitoring capabilities, Aflatoxin

Risk Index calculation, automated alerts, and user authentication. Non-functional requirements will cover system reliability, sensor accuracy, battery life, and response times. The process of directly meeting stakeholders will give us better understanding of how the technological gap can be addressed and ensure the system's requirements are precisely tailored to meet the community's real-world challenges.

3.2.2. Phase two: System Design

In this phase, we will create the blueprint for the entire SmartSilo system. We will design the system's overall architecture using Draw.io software, showing the flow of data from sensors through the communication layer to the cloud platform and finally to the user dashboard. We will also design the database schema to show how information is stored and retrieved, including sensor readings, user profiles, alert history, and system logs. Most importantly, we shall create a detailed user interface (UI) with wireframes using Figma—simple visual guides that show the layout of the application's screens, including the dashboard, real-time monitoring views, historical data charts, and alert configuration panels.

3.2.3. Phase three: Development and Implementation

This phase will focus on actively building the system's core functions through repeated development sprints, with the primary work being carried out by the project's development team.

i. **Sprint 1: Sensor Node Development** – This sprint will involve assembling and programming sensor nodes using the Arduino IDE. We will integrate DHT22 temperature and humidity sensors with ESP32 microcontrollers and LoRa modules. The nodes will be programmed to collect environmental data at regular intervals and transmit it to the gateway.

ii. **Sprint 2: Communication Layer Setup** – This sprint will focus on setting up the LoRaWAN gateway and ensuring reliable data reception. We will configure the gateway to receive transmissions from multiple sensor nodes and forward data to the cloud platform via the internet.

iii. **Sprint 3: Cloud Backend Development** – This sprint will involve developing the cloud backend using Python and the Django framework. We will create API endpoints to receive sensor data, implement the database for data storage, and develop the Aflatoxin Risk Index (ARI) algorithm in Python. The ARI algorithm will calculate real-time risk scores based on temperature and humidity conditions favorable for mold growth.

iv. **Sprint 4: Mobile app Dashboard and Integration** – This sprint will focus on developing the web dashboard using HTML, CSS, and JavaScript. We will create visualization components for displaying sensor data and ARI scores, implement the alert system (SMS/email notifications), and integrate the control logic for automated dryer triggering.

3.2.4. Phase four: Testing and Validation

Testing will be conducted continuously throughout the development process. Unit testing will be performed on individual components to ensure they function correctly, for example, verifying that sensors return accurate readings and that the ARI algorithm produces expected outputs for given inputs.

Integration testing will ensure seamless communication between all system layers: from sensor nodes to gateway, from gateway to cloud platform, and from cloud platform to mobile app dashboard and alerts.

Functional testing will verify that all technical components work correctly and reliably, checking if data is consistently transmitted, if the dashboard updates in real-time, and if alerts are triggered when risk thresholds are exceeded.

Pilot deployment will be executed in a selected grain storage facility in Gulu district for real-world testing. The participants will include at least 3 grain storage operators or cooperative managers who will use the system for a defined period. We will observe system performance, collect user feedback through surveys and interviews, and document any issues encountered.

Usability testing will be carried out with the storage operators to ensure the interface is intuitive and accessible. We will observe them without intervention to see if they can navigate the dashboard well, understand the risk indicators, and respond appropriately to alerts.

Environmental testing will evaluate system performance under actual storage conditions, including variations in temperature, humidity, and potential interference with wireless communication. Through this process, users will validate how the system performs real-world tasks, and we will identify what adjustments are needed to meet stakeholder needs.

References

Akullo, J. O., Amayo, R., Okello, D. K., Mohammed, A., Muyinda, R., Magumba, D., ... Mweetwa, A. M. (2023). Aflatoxin contamination in groundnut and maize food products in eastern and northern uganda. *Cogent Food & Agriculture*, *9*. Retrieved from `https://doi.org/10.1080/23311932.2023.2221015` doi: 10.1080/23311932.2023.2221015

Basciftci, F., Unlu, T., & Dasdemir, A. (2021). Monitoring grain silos instantly with IoT based control system. *The Eurasia Proceedings of Science, Technology, Engineering & Mathematics (EPSTEM)*, *14*, 15–23.

Doshi, J., Patel, T., & Bharti, S. K. (2019). Smart farming using IoT, a solution for optimally monitoring farming conditions. *Procedia Computer Science*, *160*, 746–751. doi: 10.1016/j.procs.2019.11.016

Dyck, I., George. (2025). *A novel digital twin framework for monitoring inventory and in-bin drying in grain storage* (Doctoral dissertation, University of Manitoba, Winnipeg, Manitoba). Retrieved from `http://hdl.handle.net/1993/39537` (MSpace Institutional Repository)

Eskola, M., Kos, G., Elliott, C. T., Hajšlová, J., Mayar, S., & Krska, R. (2020). Worldwide contamination of food-crops with mycotoxins: Validity of the widely cited 'fao estimate' of 25%. *Critical Reviews in Food Science and Nutrition*, *60*, 2773–2789. Retrieved from `https://doi.org/10.1080/10408398.2019.1658570` doi: 10.1080/10408398.2019.1658570

Hema, L. K., Velmurugan, S., Sunil, D. N., Aziz, S. T., & Thirunavkarasu, S. (2020). IOT based real-time control and monitoring system for food grain procurement and storage. *IOP Conference Series: Materials Science and Engineering*, *993*(1), 012079. doi: 10.1088/1757-899X/993/1/012079

Ndwata, A. H., Rashid, S. A., & Chaula, D. N. (2022). Aflatoxins b contamination levels in maize and awareness of aflatoxins among main maize stakeholders in chemba and kondoa districts, tanzania. *African Journal of Microbiology Research*, *16*, 223–237. Retrieved from `http://www.academicjournals.org/AJMR` doi: 10.5897/AJMR2022.9637

Pargaonkar, S. (2023). A comprehensive research analysis of software development life cycle (sdlc) agile & waterfall model advantages, disadvantages, and application suitability in software quality engineering. *Interna-*

tional Journal of Scientific and Research Publications, *13*(8), 120–124. Retrieved from http://dx.doi.org/10.29322/IJSRP.13.08.2023.p14015 doi: 10.29322/IJSRP.13.08.2023.p14015

Ramachandran, R. P., & Singh, C. B. (2022). Integrated approach on stored grain quality management with CO2 monitoring—a review. *Journal of Stored Products Research*, *96*, 101950. Retrieved from https://doi.org/10.1016/j.jspr.2022.101950 (Corrigendum published in Vol. 99 (2022), DOI: 10.1016/j.jspr.2022.102028) doi: 10.1016/j.jspr.2022.101950

Sinha, A., & Das, P. (2021). Agile methodology vs. traditional waterfall SDLC: A case study on quality assurance process in software industry. In *2021 ieee international conference on emerging technologies* (pp. 1–4). doi: 10.1109/ICET.2021.00000

Wanyonyi, E. N. (2025). A smart IoT-based real-time system for monitoring storage conditions of maize grains. *East African Journal of Information Technology*, *8*(1), 146–155. doi: 10.37284/eajit.8.1.3058

A. Questionnaire

QUESTIONNAIRE FOR GRAIN STORAGE OPERATORS / FARMERS

Title: Questionnaire on Grain Storage Practices and Aflatoxin Risk Management
Purpose: This questionnaire is intended to collect data to support the development of the Smart-Silo IoT system for monitoring and preventing aflatoxin contamination in grain storage.
Instructions: Please tick (✓) or write where appropriate. Information provided will be used strictly for academic purposes.

Section A: Respondent Background Information

1. Gender:
   ☐ Male ☐ Female

2. Age group:
   ☐ Below 20 ☐ 21–30 ☐ 31–40 ☐ 41–50 ☐ Above 50

3. Role in grain storage:
   ☐ Farmer ☐ Store manager ☐ Cooperative member ☐ Other (specify) …………………

4. Years of experience in grain storage:
   ☐ Less than 1 year ☐ 1–3 years ☐ 4–6 years ☐ Above 6 years

Section B: Grain Storage Practices

5. What type of grain do you mainly store?
   ☐ Maize ☐ Groundnuts ☐ Sorghum ☐ Beans ☐ Other …………………

6. What storage method do you currently use?
   ☐ Traditional granary ☐ Silo ☐ Bags in store ☐ Other …………………

7. How do you currently monitor grain storage conditions?
   ☐ Manual inspection
   ☐ Sun-drying
   ☐ No monitoring
   ☐ Other (specify) …………………

8. How often do you check grain storage conditions?
   ☐ Daily ☐ Weekly ☐ Monthly ☐ Rarely

Section C: Aflatoxin Awareness and Challenges

9. Are you aware of aflatoxin contamination in grains?
   ☐ Yes ☐ No

10. Have you ever experienced grain spoilage due to mold or aflatoxins?
    ☐ Yes☐ No

11. What challenges do you face in grain storage?
    ☐ High humidity
    ☐ High temperature
    ☐ Mold growth
    ☐ Lack of monitoring tools
    ☐ Other …………………

Section D: Technology and System Requirements

12. Have you ever used any technology-based system for grain monitoring?
    ☐ Yes☐ No

13. Would you be willing to use a mobile application system that monitors temperature and humidity automatically?
    ☐ Yes☐ No

14. What features would you prefer in a grain monitoring system?
    ☐ Real-time alerts
    ☐ Automated drying
    ☐ Mobile/SMS notifications
    ☐ Data reports
    ☐ All of the above

15. Any additional comments or suggestions?
    ………………………………………………………………………………………

B. Interview Guide

INTERVIEW GUIDE FOR GRAIN STORE MANAGERS AND AGRICULTURAL OFFICERS

Purpose: To gather in-depth information on grain storage challenges, aflatoxin management practices, and system requirements for the Smart-Silo project.

Interview Questions

1. Can you describe the common grain storage practices used in this facility/area?
2. What are the major challenges you face in preventing aflatoxin contamination?
3. How do you currently monitor temperature and humidity in grain storage?
4. What happens when poor storage conditions are detected?
5. Have you experienced losses due to aflatoxin contamination? Please explain.
6. What do you think are the limitations of current grain storage systems?
7. How do you feel about adopting an IoT-based automated monitoring system?
8. What features would you consider essential in a smart grain storage system?
9. What challenges do you foresee in implementing such a system?
10. Any recommendations for improving grain storage and food safety?

C. Project Budget

ESTIMATED PROJECT BUDGET FOR SMART-STUA SYSTEM

Item	Description	Quantity	Unit Cost (UGX)	Total Cost (UGX)
ESP32 mirco-controllers	IoT sensor controller	3	60,000	180,000
DHT22 sensor	Temperature & humidity	3	35,000	105,000
LoRa modules	Long-range communication	2	120,000	240,000
Power Supply	Batteries & adapters	3	40,000	120,000
Breadboards & wiring	Circuit connections	1 set	50,000	50,000
Web hosting	Cloud server & database	1	200,000	200,000
Internet data	Development & testing	1	100,000	100,000
Software Tools	Development tools (where applicable)	1	100,000	100,000
Transport	Field data collection & testing	1	180,000	180,000
Stationery	Printing, questionnaires	1	75,000	75,000
Total Estimated Cost				**1,400,000**